

Name: _____

SOLUTIONS

Date: _____



Year 9 Mathematics

Test 9, 2015

Topic – Non-linear Functions

60 %
--

Total Time: **62** minutesTotal Reading: **2** minutesTotal Working: **60** minutes

Weighting: _____ % of the year.

Equipment: Curriculum Council Formula Sheet; ½ page notes (A4 one side), CAS calculator; Scientific Calculator

SECTION 1: CALCULATOR FREE

Time: 15 minutes	Marks for Section 1: 17
Working: 15 minutes	Equipment Allowed: Nil

1. [5 marks: 1, 1, 1, 1, 1]

Determine if the following equations are linear, quadratic, hyperbola, circle or none of these

a) $3(4 - 5x) = x(2 - x^2)$

None of these

b) $(x - 4)^2 = 9 - (y + 1)^2$

Circle

c) $y - 2x = 4$

Linear

d) $y = \frac{-3}{x}$

Hyperbola

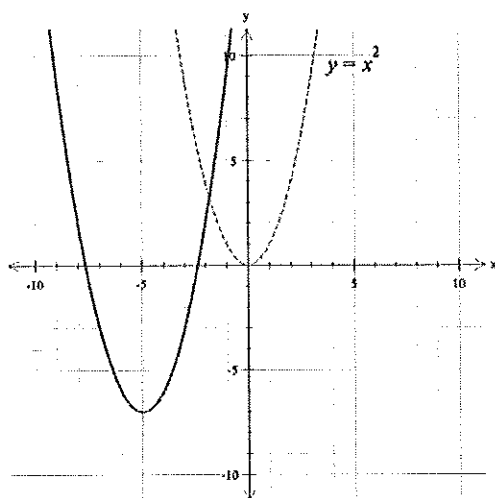
e) $2(x - 4) = x(7 + x)$

Quadratic

2. [8 marks: 3, 5]

The simplest quadratic function: $y = x^2$ is shown as a dashed line in each graph. Describe, using mathematical language, how the quadratic function with a solid line has transformed. Please note there has been no dilation.

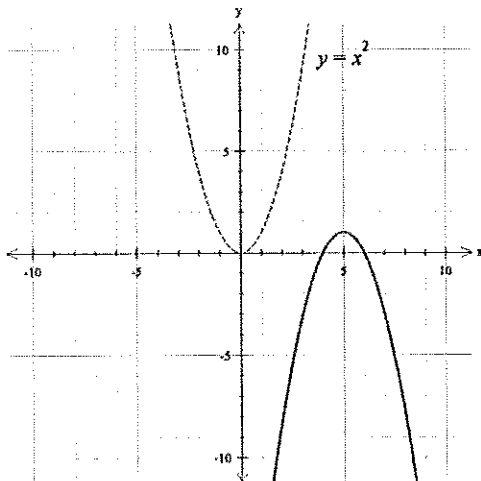
a)



Horizontal translation left
5 units

Vertical translation down
7 units

b)



Reflection over x -axis

Horizontal translation right
5 units

Vertical translation up
1 unit

3. [4 marks: 1, 1, 1, 1]

For this graph, state the following key features:

a) Axis of symmetry

$$x = 1$$

b) Nature and location of the turning point

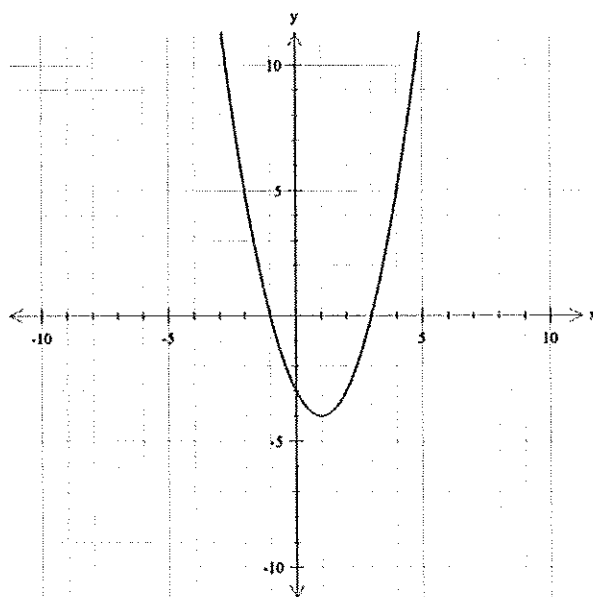
$$(1, -4) \text{ Minimum}$$

c) The coordinates of the x -intercept(s)

$$(-1, 0) \text{ and } (3, 0)$$

d) The coordinates of the y -intercept

$$(0, -3)$$



~ END OF TEST SECTION 1 ~

Name: _____

Date: _____

SECTION 2: CALCULATOR ASSUMED

Time:	45 minutes	Marks for Section 2:	43
Reading:	2 minutes	Equipment Allowed:	½ page notes (A4 one side), CAS calculator
Working:	43 minutes		

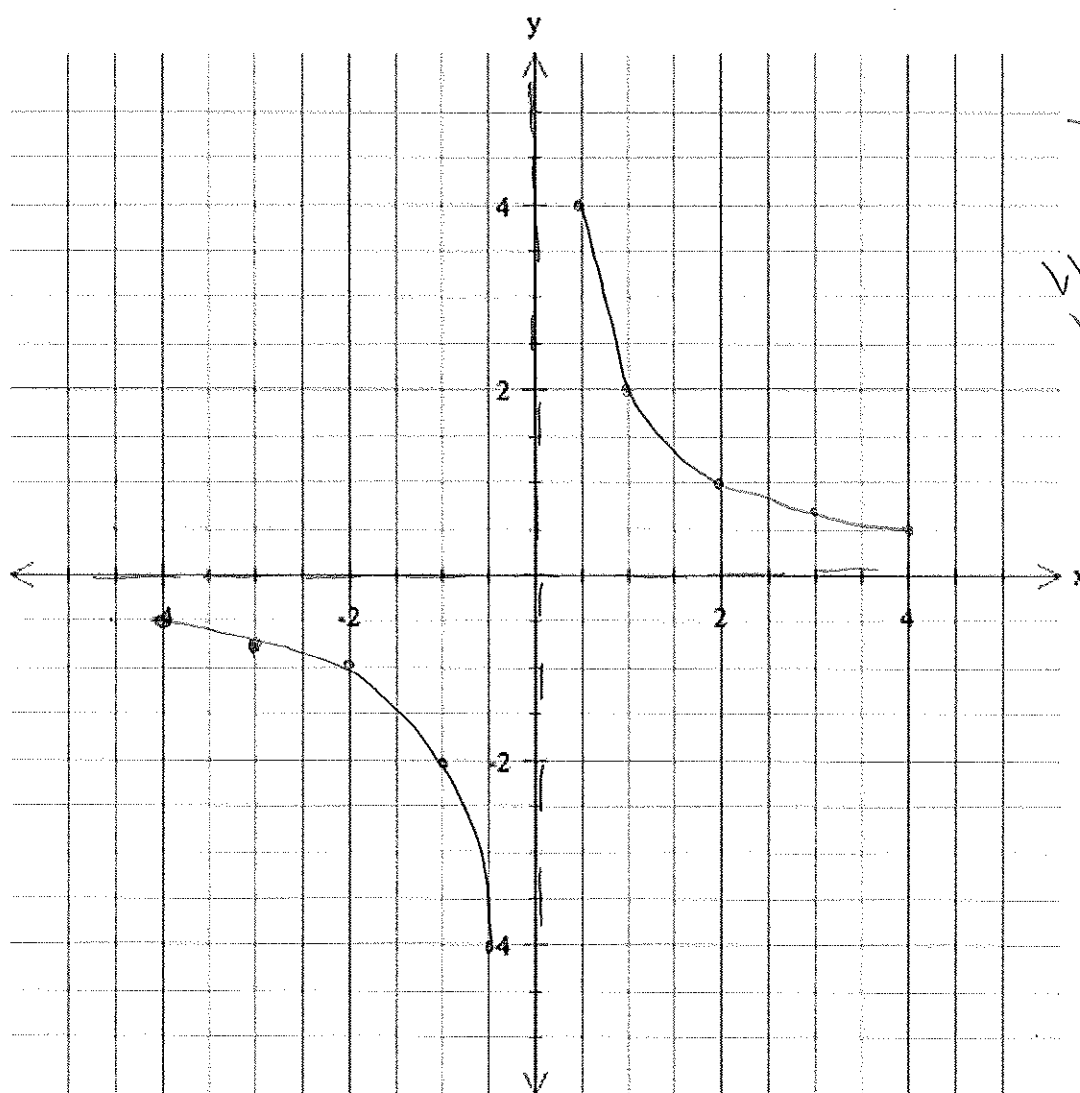
4. [5 marks]

Complete the table of values for the equation shown and plot the related graph

$$y = \frac{2}{x}$$

x	-4	-3	-2	-1	-½	0	½	1	2	3	4
y	-½	-0.7	-1	-2	-4	undef	4	2	1	0.7	½

✓✓

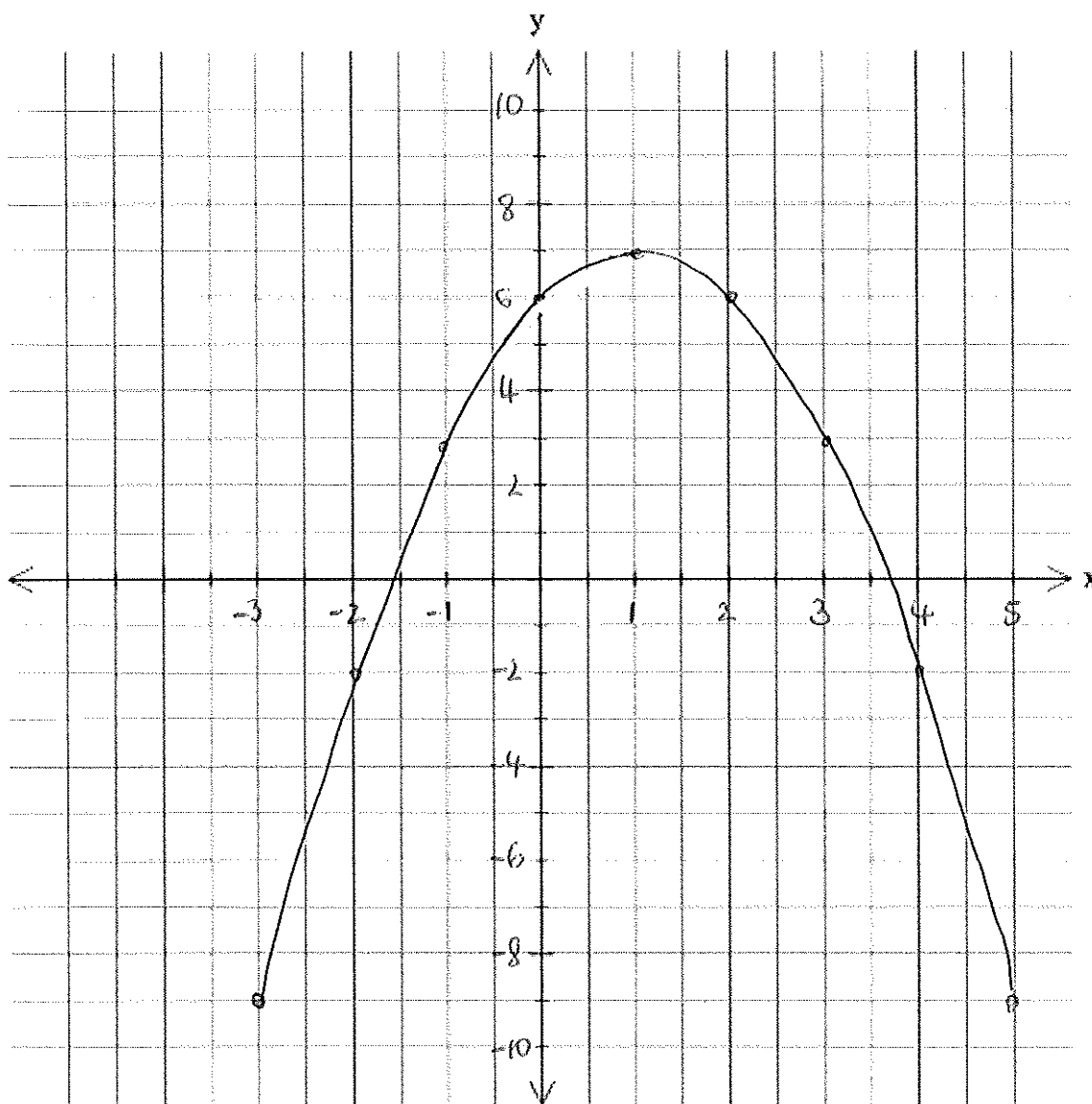
-½ no
asymptote✓✓ coordinates
✓ general

5. [5 marks]

Complete the table of values for the equation shown and plot the related graph

$$y = -(x-1)^2 + 7$$

x	-3	-2	-1	0	1	2	3	4	5
y	-9	-2	3	6	7	6	3	-2	-9



6. [6 marks: 1, 1, 1, 3]

For the equation, $y = (x + 4)^2 - 9$ determine the following:

a) The y-intercept.

$$y = (0 + 4)^2 - 9$$

$$= 16 - 9$$

b) The x-intercept(s).

$$0 = (x + 4)^2 - 9$$

$$9 = (x + 4)^2$$

$$\pm 3 = x + 4$$

$$-3 - 4 = -7$$

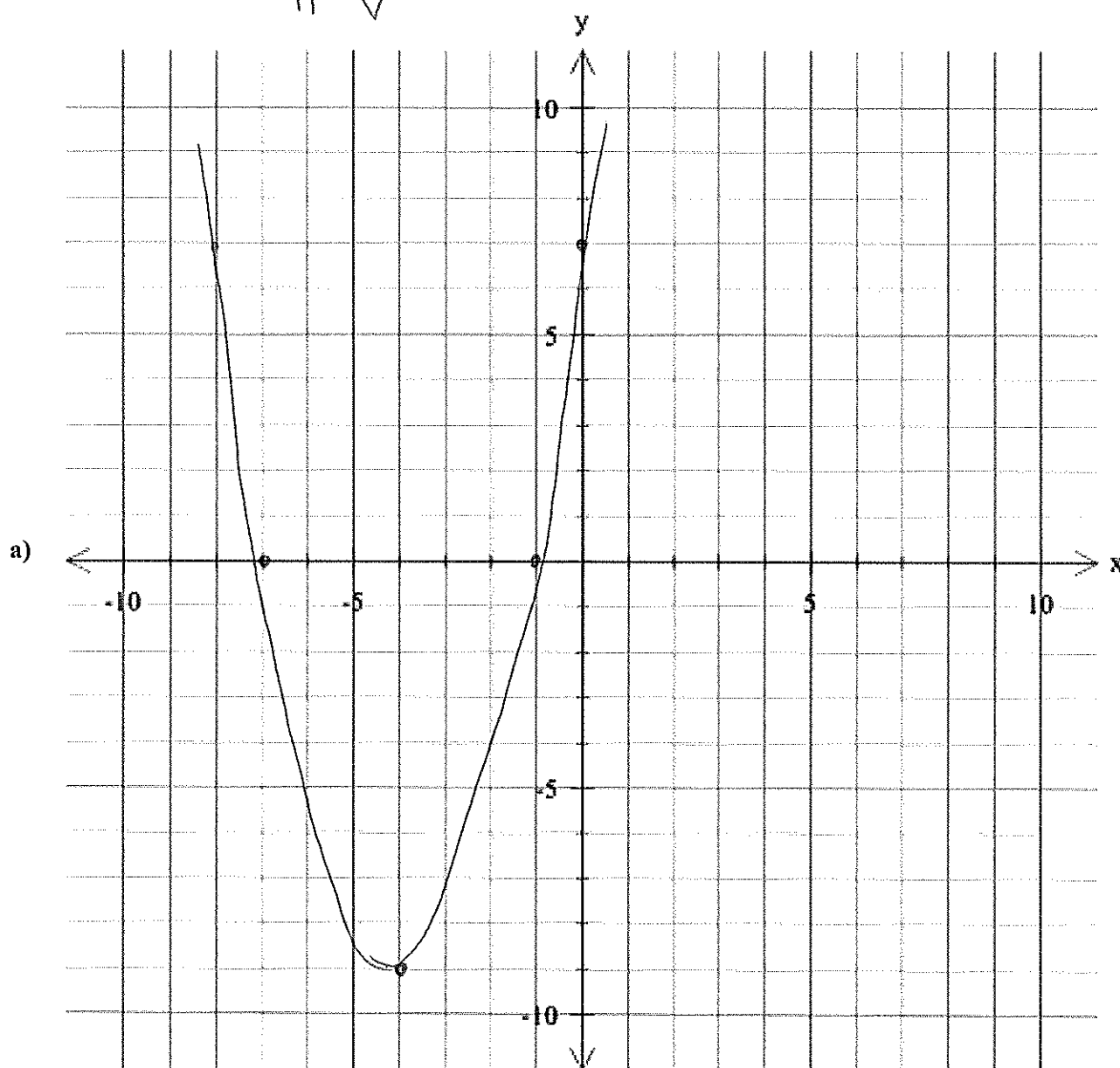
$$3 - 4 = -1$$

c) Nature and location of the turning point.

$(-4, -9)$ Minimum

d) Sketch the graph.

shape $\vee$
intercept $\vee$
TP $\vee$

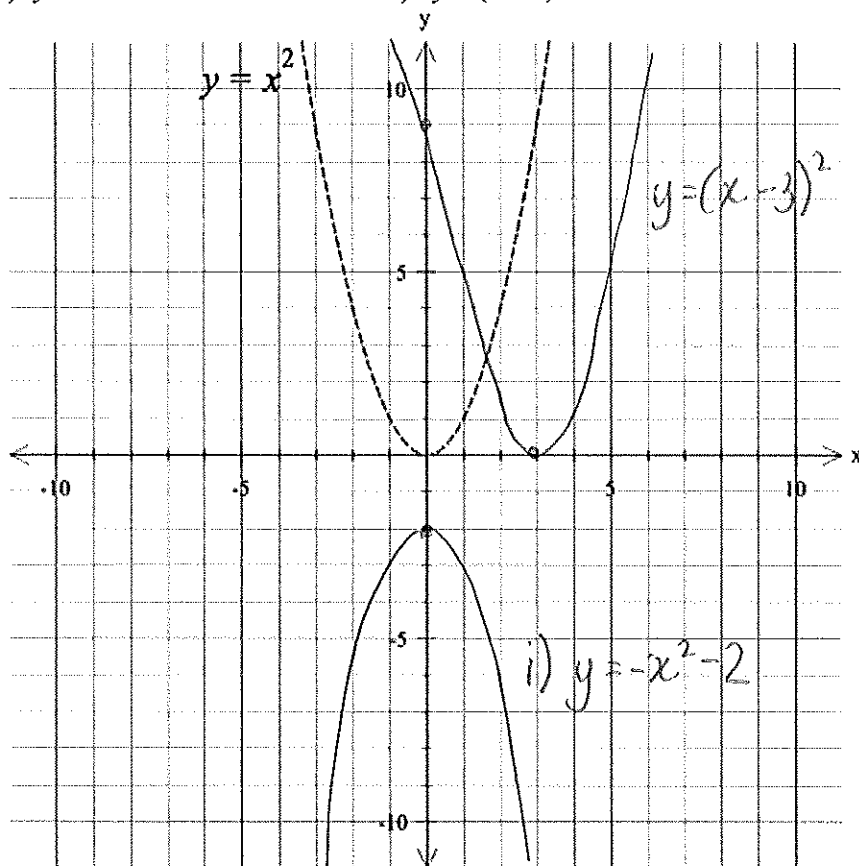


7. [8 marks: 4, 4]

The grid below shows the function $y = x^2$. On the same grid sketch the following functions (including the correct y-intercept) and label each function clearly.

a) i) $y = -x^2 - 2$

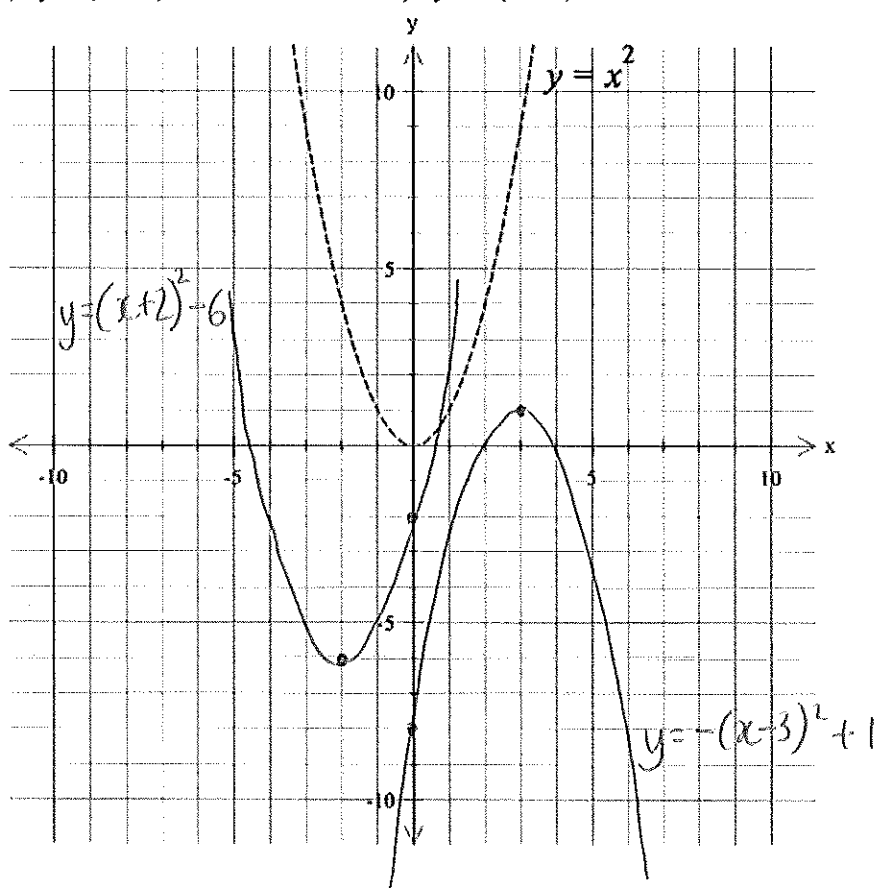
ii) $y = (x-3)^2$



$$\begin{aligned} y &= (x-3)^2 \\ &= (0-3)^2 \\ &= 9 \end{aligned}$$

b) i) $y = (x+2)^2 - 6$

ii) $y = -(x-3)^2 + 1$



$$\begin{aligned} y &= (0+2)^2 - 6 \\ &= 4 - 6 \\ &= -2 \end{aligned}$$

$$\begin{aligned} y &= -(0-3)^2 + 1 \\ &= -9 + 1 \\ &= -8 \end{aligned}$$

8. [6 marks: 1, 1, 1, 1, 1, 1]

Match each of the following equations to one of the graphs below:

a) $y = -5x^2$ (C)

b) $y = x^2 + 4$ (D)

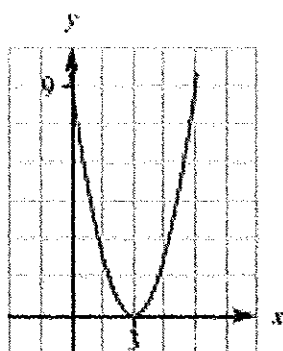
c) $y = (x - 2)^2 + 3$ (B)

d) $y = (x - 3)^2$ (A)

e) $y = (x + 2)^2$ (F)

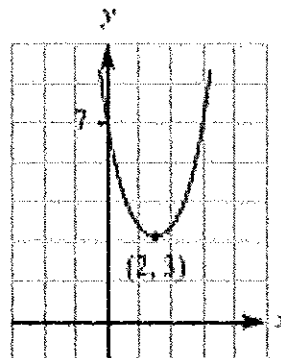
f) $y = \frac{1}{4}x^2$ (E)

A



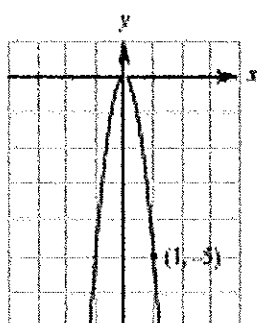
Equation: $y = (x - 3)^2$
(d)

B



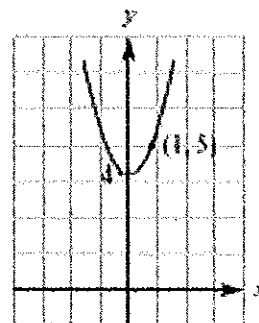
Equation: $y = (x - 2)^2 + 3$
(c)

C



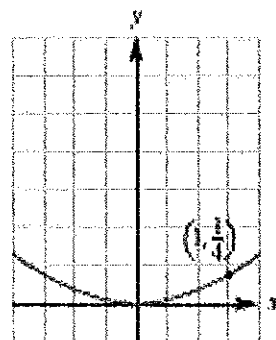
Equation: $y = -5x^2$
(a)

D



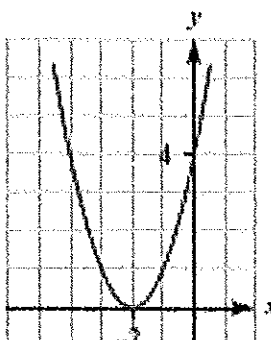
Equation: $y = x^2 + 4$
(b)

E



Equation: $y = \frac{1}{4}x^2$
(f)

F



Equation: $y = (x + 2)^2$
(e)

9. [1 mark]

Which graph matches the equation: $(x+5)^2 + (y+6)^2 = 9$

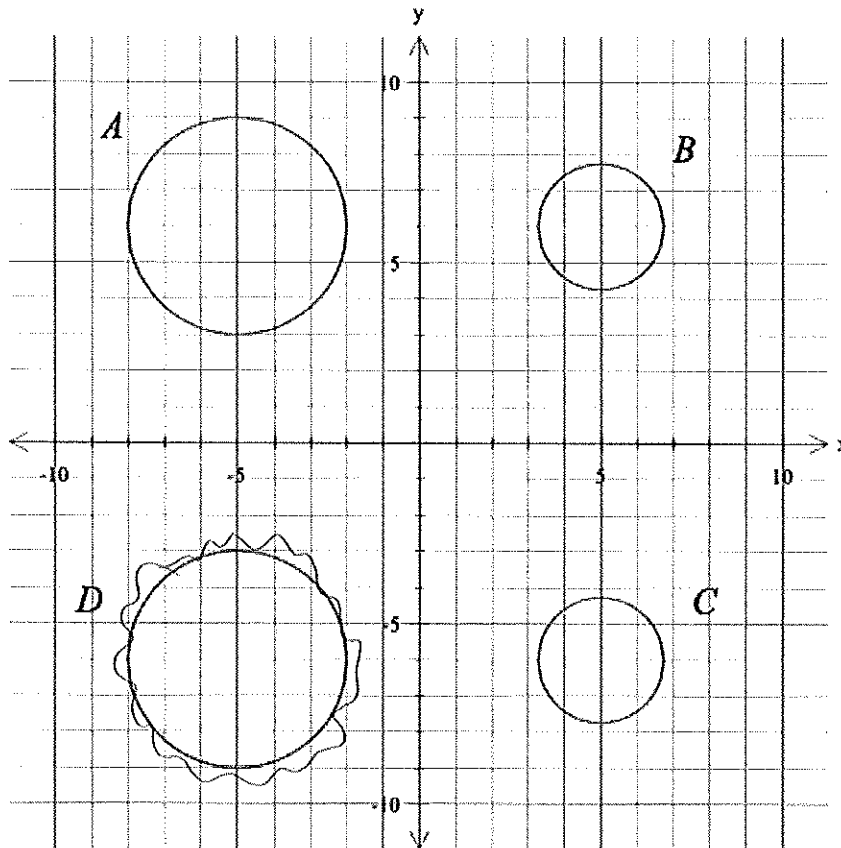
GRAPH:

A

B

C

D



10. [2 marks: 1, 1]

Given the equation: $(x-1)^2 + (y+3)^2 = 4$

a) What is the coordinate of the centre of the graph

$(1, -3)$

b) What is the radius of the circle

2

11. [4 marks: 1, 1, 1, 1]

Factorise the following quadratic equations:

a) $x^2 - 25$

$$= x^2 - 5^2$$

$$= (x+5)(x-5)$$

b) $49a^2 - 4b^2$

$$= (7a)^2 - (2b)^2$$

$$= (7a+2b)(7a-2b)$$

c) $(x-3)^2 - 49$

$$= (x-3)^2 - 7^2$$

$$= (x-3+7)(x-3-7)$$

$$= (x+4)(x-10)$$

d) $x^2 - 6x + 8$

$$= (x-4)(x-2)$$

12. [6 marks: 1, 1, 1, 1, 1, 1]

Solve the following quadratic equations:

a) $(x-1)(x+4) = 0$

$$\begin{array}{cc} \downarrow & \downarrow \\ x-1=0 & x+4=0 \\ x=1 & x=-4 \end{array}$$

b) $3x(2x+5) = 0$

$$\begin{array}{cc} \downarrow & \downarrow \\ 3x=0 & 2x+5=0 \\ x=0 & 2x=-5 \\ & x=-\frac{5}{2} \end{array}$$

c) $(x-17)^2 = 0$

$$(x-17)(x-17) = 0$$

$$\begin{array}{cc} \downarrow & \downarrow \\ x-17=0 & x-17=0 \\ x=17 & x=17 \end{array}$$

d) $3x^2 - 12 = 0$

$$3x^2 = 12 \quad (\div 3)$$

$$x^2 = 4 \quad (\sqrt{})$$

$$x = \sqrt{4}$$

$$x = \pm 2$$

e) $x^2 + x - 12 = 0$

$$(x+4)(x-3) = 0$$

$$\begin{array}{cc} \downarrow & \downarrow \\ x+4=0 & x-3=0 \\ x=-4 & x=3 \end{array}$$

f) $x^2 - 5x - 24 = 0$

$$(x-8)(x+3) = 0$$

$$\begin{array}{cc} \downarrow & \downarrow \\ x-8=0 & x+3=0 \\ x=8 & x=-3 \end{array}$$

~ END OF TEST SECTION 2 ~